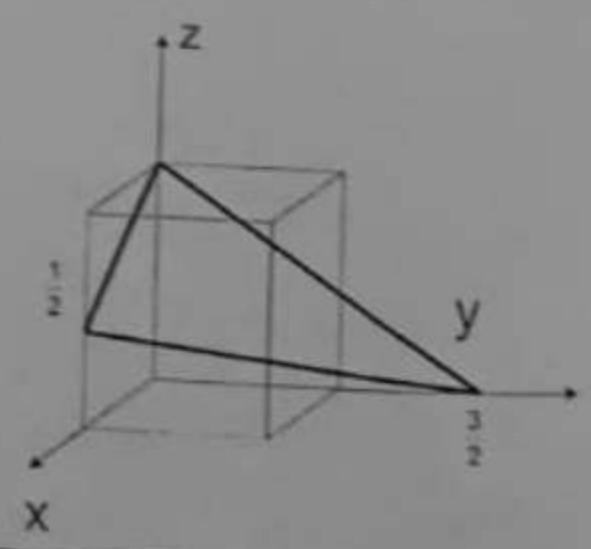
(Please write your Enrollment Number) Enrollment No. 02701222022

MID-TERM EXAMINATION

(Course Name:B.Tech CSE/ECE) (Semester: Second)

(May, 2023)OFF LINE mode

Subject Code: BAS-104	Subject: APPLIED PHYSICS-II
Time: 1 ½Hours	Maximum Marks: 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) State Gauss's Law. Derive the differential form of Gauss's Law.		
	(b) In Free space, the electric field is given as $\vec{E} = 20\cos[\omega t - 50x]\hat{y}$ V/m Calculate (i) Angular frequency, ω (ii) Displacement Current, J_d		
	(c) Explain Bloch theorem for one dimensional periodic potential.		
	(d) The number density of conduction electrons is equal to $18.1 \times 10^{28} m^{-3}$ in aluminium. Calculate its Fermi Energy? Use: Planck's Constant, $h = 6.626 \times 10^{-34}\ Js$ Mass of electron, $m_e = 9.1 \times 10^{-31}\ kg$		
Q2	(Attempt any Two Parts)		(5,5)
	(a) Using Maxwell's Equations, show that the velocity of EM waves in a dielectric medium will always be less than the velocity of light.		
	(b) Derive the equation of continuity. Explain how Maxwell corrected and presented the Ampere's law in its generalized form.		
	(c)A plane electromagnetic wave propagating in free space has magnetic field $\vec{B}(x,y,z,t) = B_o \sin\left[(x+y)\frac{k}{\sqrt{2}} + \omega t\right]\hat{k}$ Where k is the wave number and$\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are Cartesian unit vectors in x, y and z directions respectively. Find the Electric field $\vec{E}(x,y,z,t)$.		
Q3	(Attempt any Two Parts)		(5,5)
	(a) Discuss the Kronig-Penney model for the motion of electrons in a periodic potential. Explain the formation of allowed and forbidden energy bands using the $E-K$ Diagram.		
	(b) What are Miller Indices? Give their significance. (i) Draw the crystal planes corresponding to the miller indices $(\bar{2}\,20)$ and (321). (ii) Assuming a simple cubic lattice, determine the Miller Indices of the plane given below.  		
	(c) A Bravais lattice is specified by the primitive lattice vectors: $\vec{a} = a\,\hat{x}, \qquad \vec{b} = \frac{1}{2}a\,\hat{x} + \frac{\sqrt{3}}{2}a\,\hat{y}, \qquad \vec{c} = 2a\,\hat{k}$ Where $\hat{x}$, $\hat{y}$ and $\hat{z}$ are the unit vectors along the x, y, and z axes respectively, and a is the lattice distance. Find the primitive reciprocal lattice vectors $\vec{a}^*$, $\vec{b}^*$ and$\vec{c}^*$.		